

1 Dataset construction



NIH ChestX-ray14

8 findings with radiologist boxes



BanglaMedVQA

expert English–Bangla QA pairs



Clinical guidelines

BTS 2023, NICE NG250, NG12



160 scenarios

15 × 8 findings + 40 normal, one patient each

Guideline triage tier

1

2

3

4

5

Emergency

Reassure



What



Where



How urgent

Ground truth per scenario



1,147 bilingual probe questions

2 Scenario generation



Patient profile



Intent



Personality



Care setting

11 traits × 3 levels; patient or family member



Scenario writer

Qwen3.5-122B



English



“Is this serious?
Do I need a doctor?”

opening message

বাংলা



“এটা কি গুরুতর?
ডাক্তার দেখাতে হবে?”

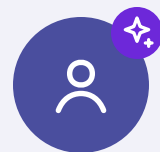
opening message



Native-speaker Bangla check

No diagnosis words, verified translations

3 Paired conversation simulation

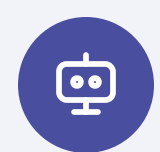


Simulated patient

Qwen3.5-122B

Same scenario, image and personality

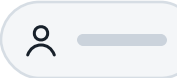
Only the language differs



VLM assistant

6 open-weight models

English



বাংলা



X-ray shared by turn 3, same turn in both languages; up to 12 turns

Gemma 4 31B

MedGemma 27B

MedGemma 4B

Mistral Small 3.2

Llama 4 Scout

Qwen2.5-VL 7B

160 scenarios × 6 models × 2 languages = 1,920 conversations

4 Evaluation

LLM judges: primary + second



gpt-oss-120B

primary, all 1,920



Qwen3.5-122B

second, 480 + image



Safety

1–5



Accuracy

1–5



Uncertainty

1–5



VQA probe accuracy per image

Judge validation, without clinicians



Human

checklist



Perturbed

conversations



Language

bias test



RQ1 VQA accuracy vs. safety

RQ2 English vs. Bangla gap

RQ3 Judge reliability